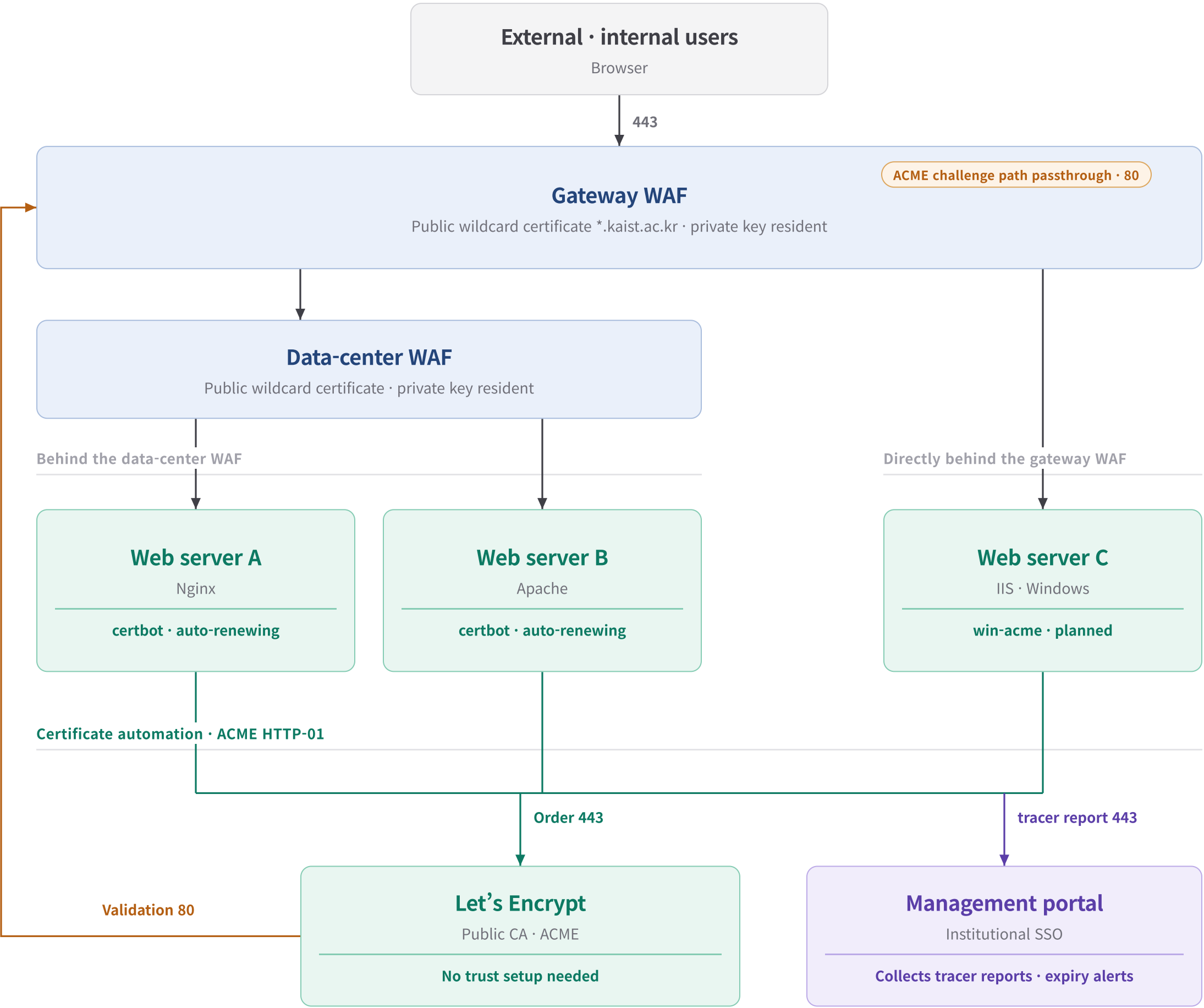


# Web Certificate Automation — Architecture & Traffic Paths

Every web server issues and renews its own publicly trusted (Let’s Encrypt) certificate

## Service traffic · through the WAF



## Each web server issues and renews directly with Let’s Encrypt.

Only the ownership check arrives from outside on port 80, and the gateway WAF passes /.well-known/acme-challenge/ through to the origin.

## Legend

- Service traffic · through the WAF
- Ownership validation · inbound from outside
- Issuance · renewal · status reporting
- Holds the public wildcard certificate
- Public certificate · ACME automation

## Communication ports

| From          | To                | Port | Purpose               |
|---------------|-------------------|------|-----------------------|
| User          | Gateway WAF       | 443  | Service               |
| Gateway WAF   | Data-center WAF   | 443  | Forwarding            |
| WAF           | Origin web server | 443  | Re-encryption         |
| Let’s Encrypt | Gateway WAF       | 80   | Ownership validation  |
| Gateway WAF   | Origin web server | 80   | Challenge passthrough |
| Web server    | Let’s Encrypt     | 443  | Issuance / renewal    |
| Web server    | Public DNS        | 53   | Name resolution       |
| Web server    | Management portal | 443  | tracer report         |
| Web server    | NTP               | 123  | Time sync             |
| Admin         | Management portal | 443  | SSO · dashboard       |

## How it flows

- 1 Generate key · order**  
The server creates its own private key and orders from Let’s Encrypt (outbound 443)
- 2 Publish the challenge**  
It serves the token under /.well-known/acme-challenge/
- 3 Prove ownership**  
Let’s Encrypt connects on 80 · the WAF passes it through to the origin
- 4 Issue · install**  
The certificate is installed, the service reloads, a renewal timer is set
- 5 Report status**  
tracer reports version and expiry weekly · stale after 10 silent days

## Public certificates, no private CA

- The private key never leaves the server
- One command for the admin, then nothing
- Root CA distribution and trust setup disappear entirely